



V93000 Specifications

V93000 Wave Scale RF-6 RFIM-6 RFIM-8

Version 8.6.0

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Notices

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⚠ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

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Wave Scale RF-6 with RFIM-6 or RFIM-8 specifications

Scope of specifications

The Advantest V93000 specifications are valid at the device with an adequate DUT board and valid calibration, i.e. system calibration (see section "Calibration and Operation", below) and fixture delay measurement (TDR). Since design and manufacturing of the DUT board are beyond Advantest's control, Advantest specifies and verifies the specifications up to the pogo level. All specifications assume a 50 Ω resistive source or load.

Specifications describe warranted product performance.

Characteristics, included as typical values, provide additional useful information, but do not represent warranted performance.

These specifications are valid for recommended system configurations.

These specifications are applicable for SmarTest 8.6.0.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

Calibration and operation

Warm up time	60 minutes
Calibration period	3 months
RF interface module activation lifetime	12 months after first activation
RF performance check	Optional: You can run it after system installation, after a repair or if any issue is suspected.

The RFIM activation lifetime is the elapsed time between the first activation and the need to have this RFIM exchanged by another one that has been factory calibrated.

RFIM calibration can only be done at Advantest factory. It cannot be done at customer site.

The external calibration and external adjustment period for a newly shipped system begins on the date of activation at your site, and not on the factory calibration and adjustment date disclosed on the Certificate of Calibration (CoC). As long as the shelf life (one year) is not expired at date of activation, the factory calibration and adjustment is still valid for the subsequent activation lifetime.

Valid at ambient temperature within ± 2.5 K of calibration temperature. For temperature requirements during calibration, please refer to the "Maintenance Guide".

WSRF calibration is required to meet the specifications outlined in this document.

WS system calibration with SmarTest 8.0.4 (or higher)

WSRF system calibration interval	Stim	Measure
3 months	0.2dB (typ) off vs calibration	0.15dB (typ) off vs calibration

Environmental

Operating temperature	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	<80% R.H., non condensing
Storage (without water)	-40°C to +70°C

Compatibility

Wave Scale RF can be installed in a slot powered by universal DC/DC power units (E8003AA). It is compatible with Smart Scale generation or later digital cards and power supplies for example, PS1600, PS9G, DPS64/128) but can also be used without any other type of card.

System software

- Wave Scale RF requires SmarTest release 8.0.2 or higher
- Linux Red Hat Enterprise 7 or later releases supported by SmarTest

Wave Scale RF-6 with RFIM-6 or RFIM-8 stimulus

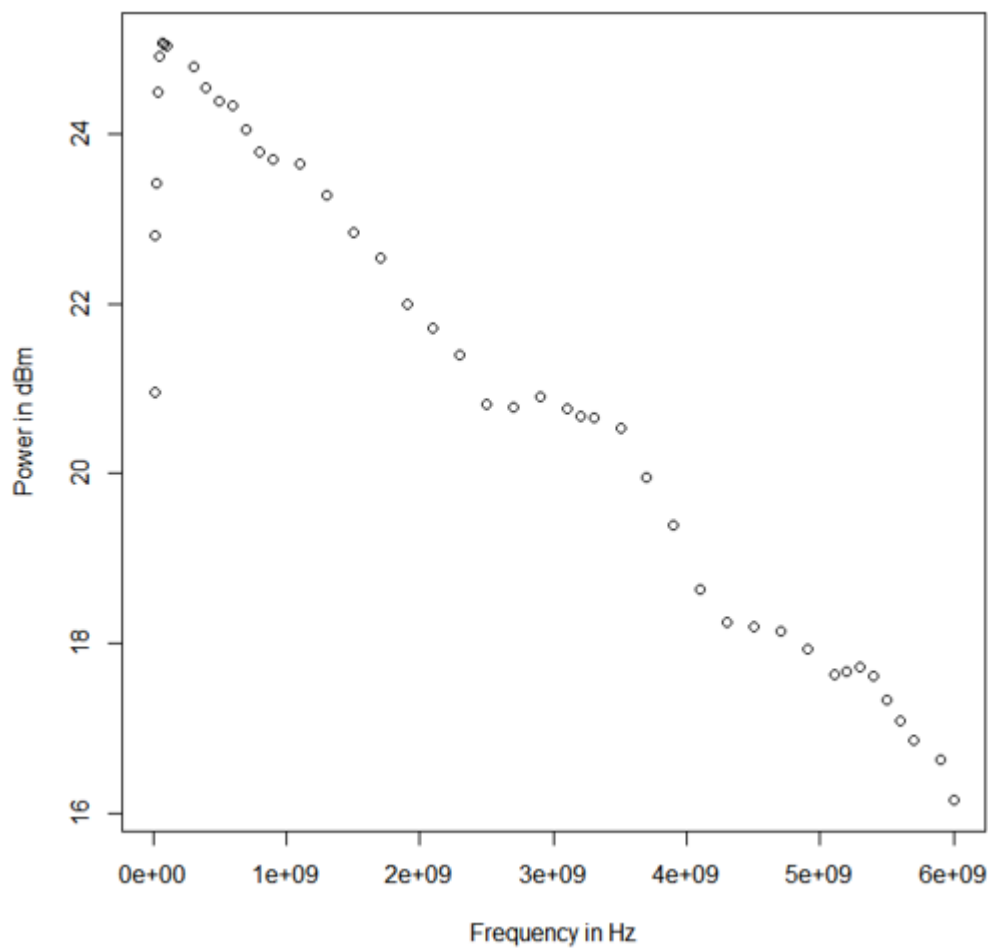
RFIM-6 stimulus maximum output power

Shows absolute maximum output power, measured as saturation power.

It is not recommended to use the Wave Scale Precision Source for power levels less than -80 dBm.

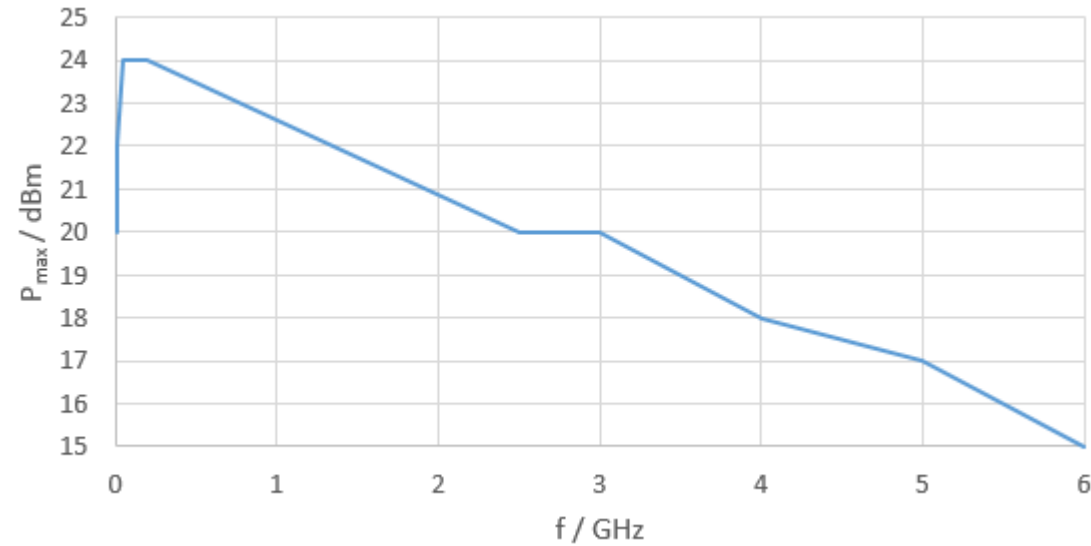
Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
0.011	17	20.5
0.016	18	22.5
0.031	21	24
0.051	21.5	24.5
0.1	21.5	24.5
1	21	23
2	18	21.5
2.5	18	20.8
3	17	20.5
4	14	18.5
5	13	17.5
6	11.5	15.5

Wave Scale RF stimulus typical maximum power



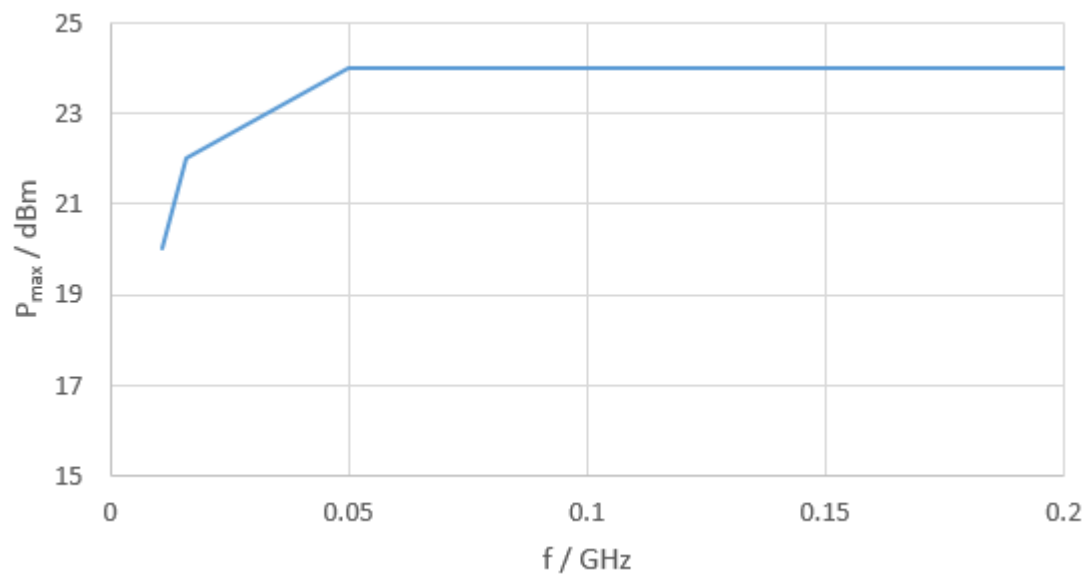
Wave Scale RF stimulus typical maximum power

Wave Scale RF stimulus typical maximum output power at 0-6 GHz



Wave Scale RF stimulus typical saturated power at 0-6 GHz

Wave Scale RF stimulus typical maximum output power at 0-0.2 GHz



Wave Scale RF stimulus typical saturated power at 0-0.2 GHz

RFIM-6 stimulus modulation flatness

The measured performance for Wave Scale RF stimulus modulation flatness is shown in the table below.

Carrier frequency range (GHz)	Maximum modulation bandwidth (MHz)	10 MHz	20 MHz	40 MHz	90 MHz	100 MHz	200 MHz
0.06 - 0.1	10	±0.15					
0.1 - 0.96	90	±0.15	±0.3	±0.5	±1.2		
0.96 - 3	200	±0.1	±0.2	±0.3		±0.8	±1.3
3 - 6	200	±0.1	±0.2	±0.4		±1.2	±1.7

RFIM-6 stimulus accuracy

Stimulus power accuracy conditions:

- Standard deviation index k=3 (coverage factor of expanded uncertainty) unless specified differently.
- ±0.5K relative water temperature stability after calibration.
- All power levels are CW signals in a 50Ω system unless noted.

Stimulus power accuracy

Power range (dBm)	-110...	-100...	-90...	-80...	-70...	-60...	-50...	-40...	-30...	-20...	-10...	0...5
Frequency range (GHz)	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...	0.05...
	±0.81	±0.70	±0.54	±0.48	±0.48	±0.48	±0.48	±0.48	±0.48	±0.48	±0.48	±0.51
	1...2	±0.71	±0.63	±0.55	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49
	2...3	±0.72	±0.72	±0.64	±0.57	±0.57	±0.57	±0.52	±0.52	±0.52	±0.52	±0.52

3...4	± 1.8 [1]	± 1.18	± 0.82	± 0.66	± 0.66	± 0.59	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53
4...5	± 1.8 [1]	± 1.18 [1]	± 0.82	± 0.66	± 0.66	± 0.59	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53
5...6	± 1.8 [1]	± 1.19 [1]	± 0.84	± 0.68	± 0.68	± 0.68	± 0.57	± 0.57	± 0.57	± 0.57	± 0.57	± 0.57

RFIM-6 stimulus tone-combiner TOI (third-order intercept)

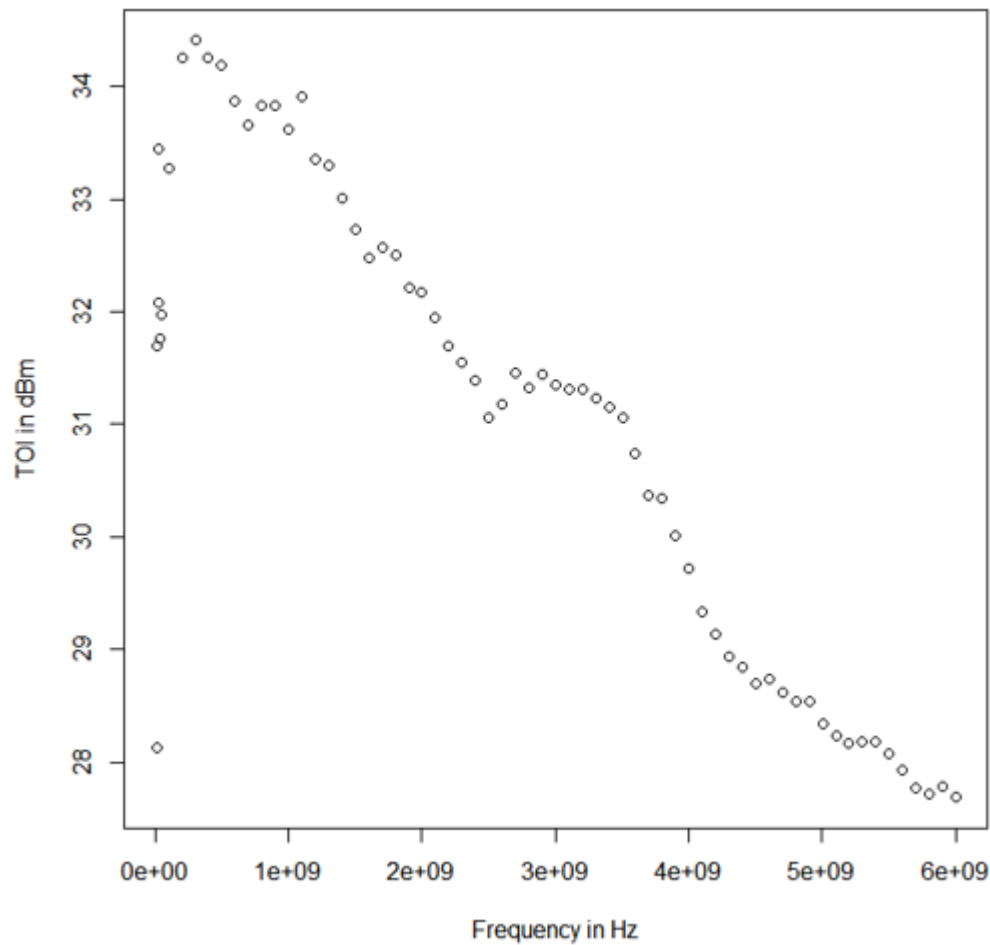
Conditions ^[2]: Tone-combiner mode, 100kHz tone spacing, optimizer balanced, 10 kHz Noise-Bandwidth.

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.1	0	28	32.5
0.2	0	29	34
0.5	0	29	34
1	0	29	34
2	0	28	32
3	0	27	31.5
4	0	25	30
5	0	24	28
6	0	23	27.5

^[1] Standard deviation index $k=2$ (Typical)

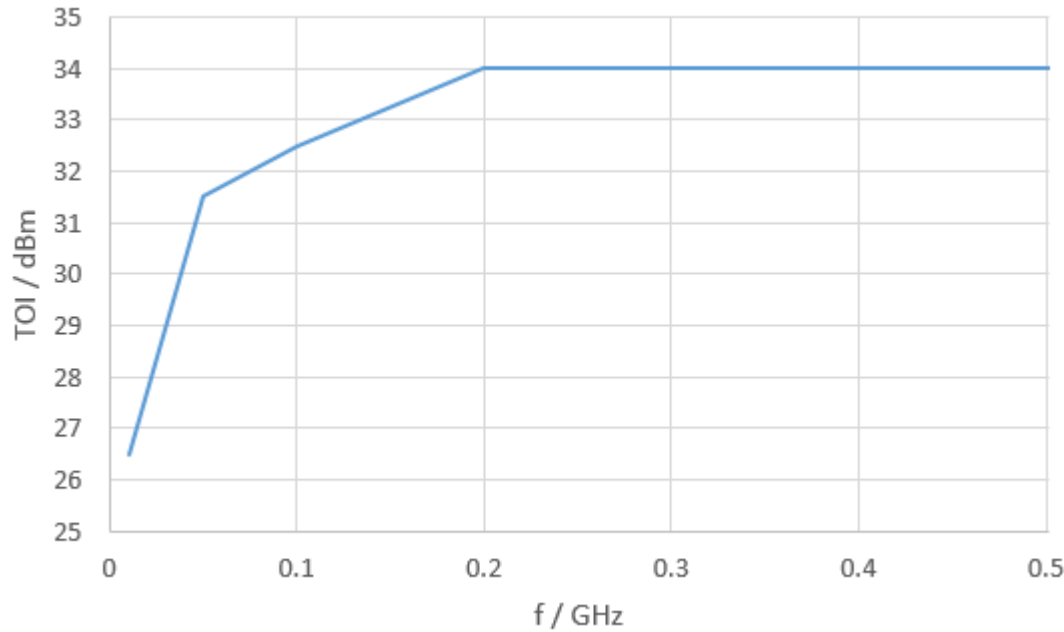
^[2] The maximum power delta between the two tones that the WSRF-6 card can guarantee is 60 dB for the above conditions.

Wave Scale RF stimulus typical TOI tone-combiner mode



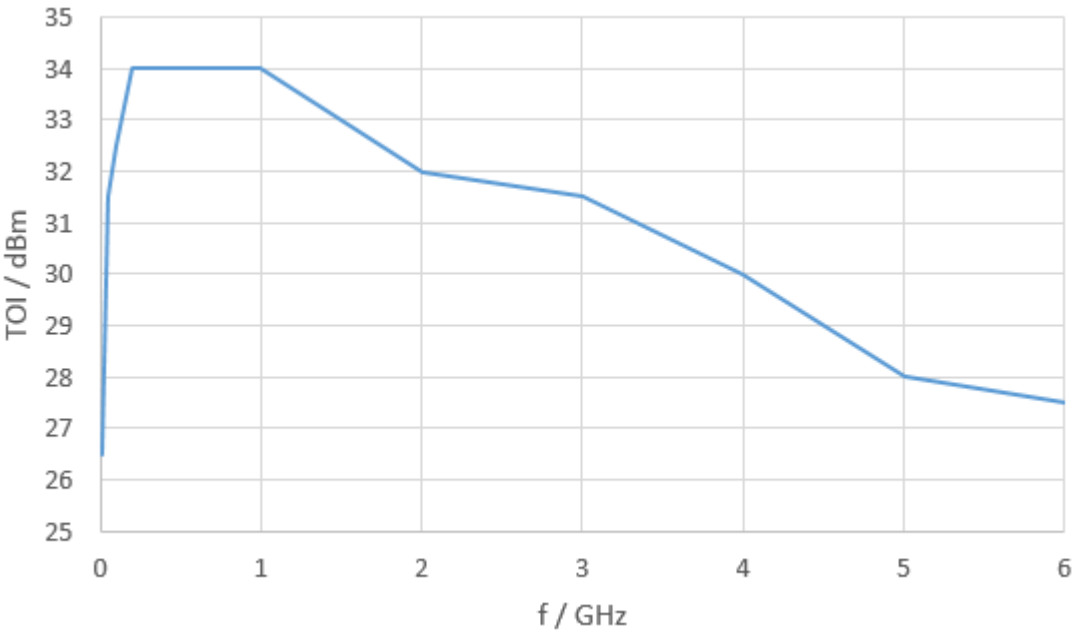
Wave Scale RF stimulus typical TOI tone-combiner mode

Wave Scale RF stimulus TOI tone-combiner typical performance 0-0.5 GHz



Wave Scale RF stimulus TOI tone-combiner typical performance 0-0.5 GHz

Wave Scale RF stimulus TOI tone-combiner typical performance 0-6 GHz



Wave Scale RF stimulus TOI tone-combiner typical performance 0-6 GHz

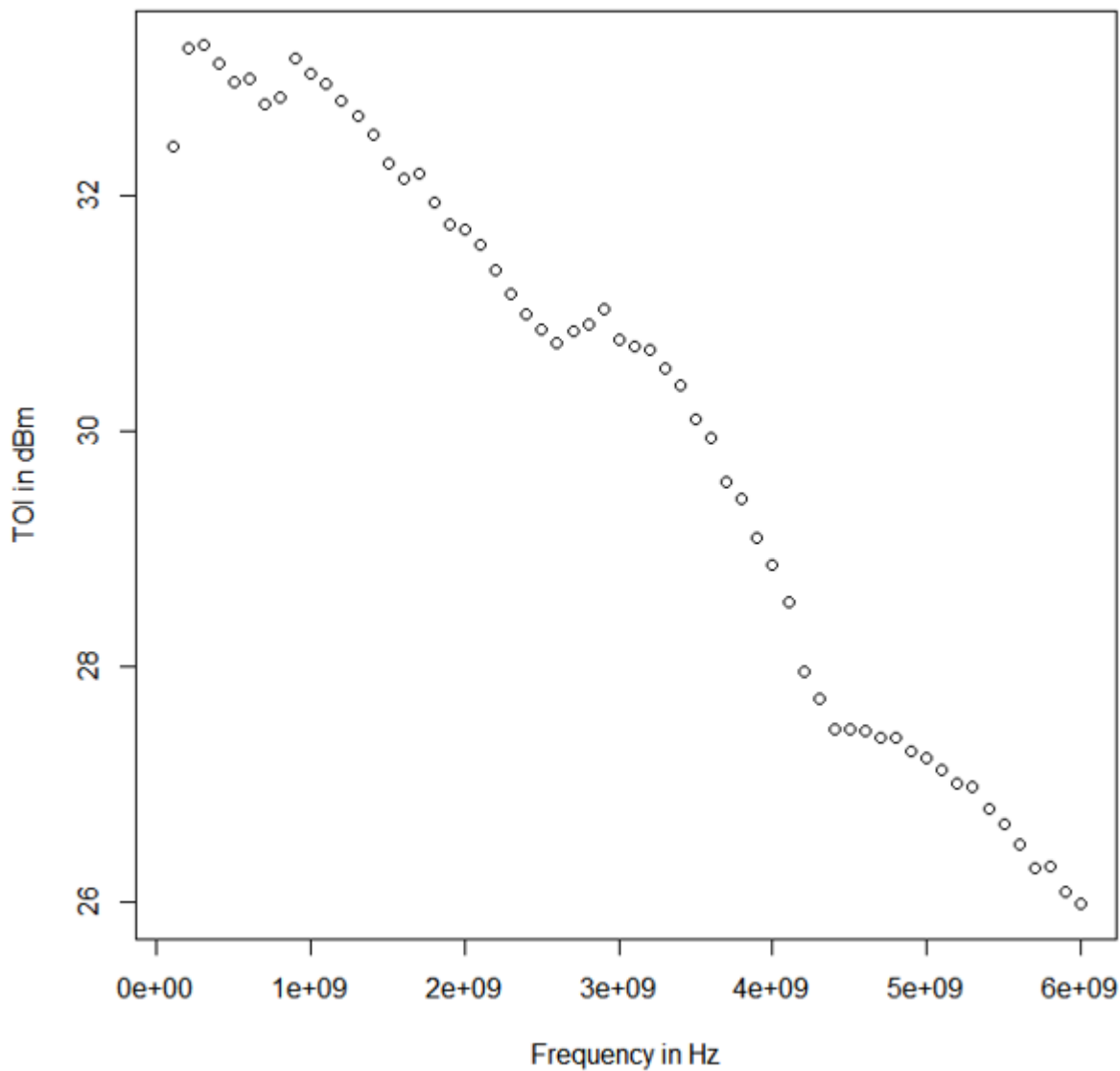
RFIM-6 stimulus modulated TOI (third-order intercept)

IQ-modulated 2-tone signal, tone-spacing 100kHz ^[3]

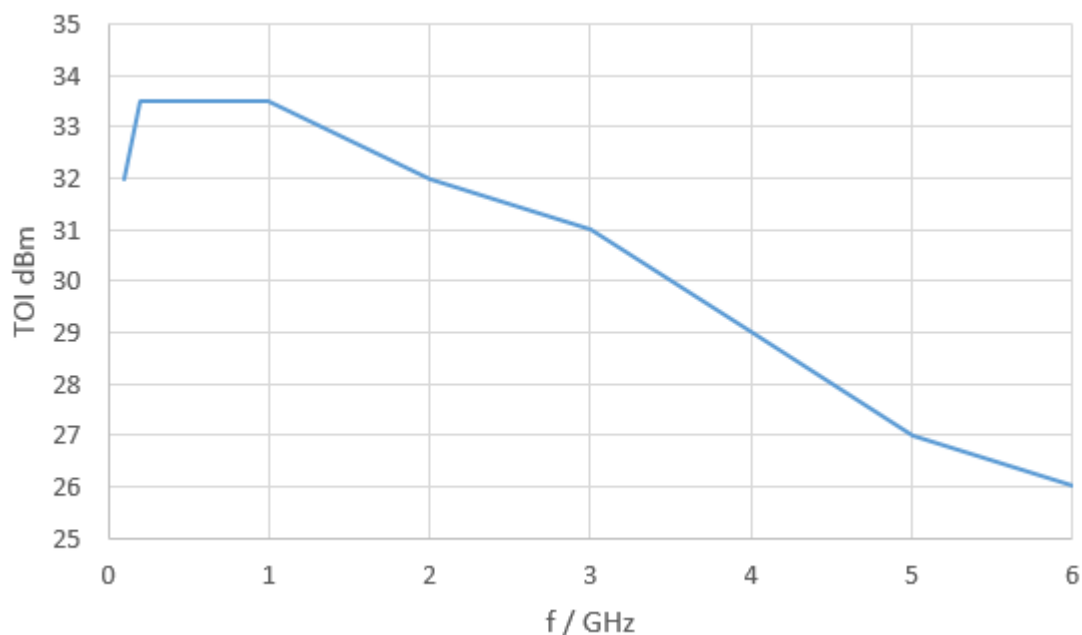
Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.5	0	28.5	33.5
1	0	28.5	33.5
2	0	27.5	32
3	0	27	31
4	0	24	29
5	0	22.5	27
6	0	21.5	26

^[3] 2-tone IQ Signal, 250Msps, (tones at -10MHz, -10.1MHz, 0dbm each tone power), total power +3dBm, optimizer balanced, 10kHz Noise-Bandwidth

Wave Scale RF stimulus typical TOI modulated



Wave Scale RF stimulus typical TOI modulated

Wave Scale RF stimulus modulated TOI (third-order intercept) at 0-6 GHz

Wave Scale RF stimulus modulated TOI (third-order intercept) at 0-6 GHz

RFIM-6 stimulus harmonics

Carrier frequency range (GHz)	Output power (dBm)	Harmonics, 2nd and above (dBc)	Subharmonics, 2nd (dBc)	Subharmonics, 3rd (dBc)
0.01 - 6	≤ 0	< -30	< -30	< -30

RFIM-6 stimulus phase noise

Specified CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power.

WSRF Stimulus Phase Noise (dBc/Hz)

Frequency [MHz]	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
50	-90	-104	-114	-116	-132
320	-90	-104	-114	-116	-132
545	-103	-117	-126	-130	-143
600	-103	-117	-126	-130	-143
885	-99	-113	-124	-127	-142
1900	-92	-107	-117	-119	-138
2400	-90	-106	-116	-119	-136
5800	-82	-98	-108	-111	-129

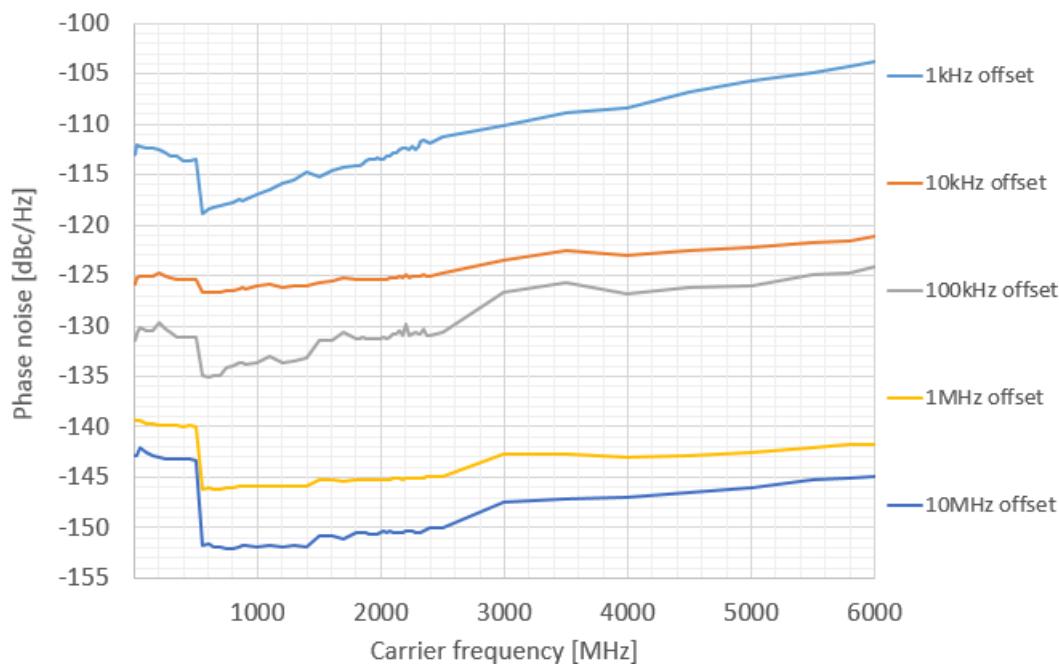
WSRF + WSPS stimulus phase noise max [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-112	-121	-129	-142	-148
600	-112	-121	-129	-142	-148
885	-111	-120	-128	-142	-148
1900	-108	-119	-125	-141	-147
2400	-105	-119	-125	-141	-147
5800	-95	-115	-117	-138	-141

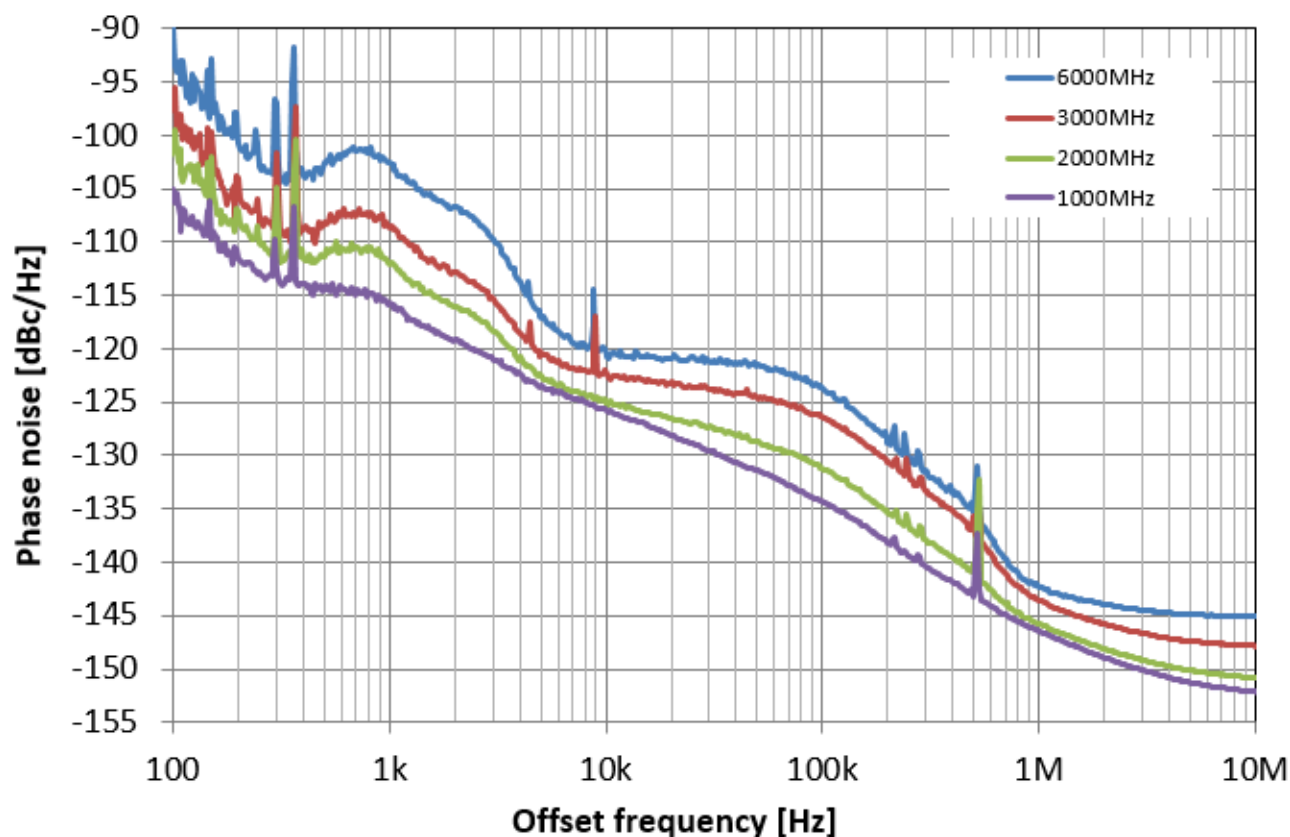
WSRF + WSPS stimulus phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-111	-124	-130	-139	-141
320	-111	-125	-130	-139	-143
545	-118	-126	-134	-145	-151
600	-117	-126	-134	-145	-151
885	-116	-125	-133	-145	-151
1900	-112	-125	-131	-144	-150
2400	-110	-124	-130	-144	-149
5800	-103	-121	-124	-141	-144

The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus carrier frequency for several offset frequencies.



The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus offset frequency for several carrier frequencies.



RFIM-6 stimulus random jitter

Measured at 0 dBm stimulus power.

WSRF Stimulus random jitter

Frequency (MHz)	RMS jitter bandwidth	Random jitter (fs_{rms}) Typ	Random jitter (fs_{rms}) Max
600-6000	10 kHz to 10 MHz	110	155

Integration bandwidth : 10kHz to 10MHz

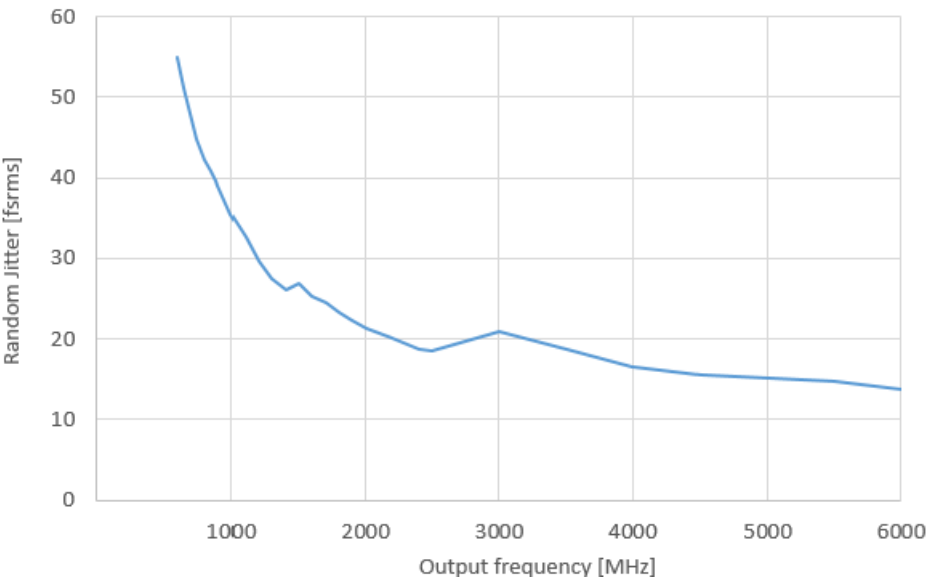
WSRF + WSPS stimulus random jitter max

Frequency [MHz]	Random jitter max [fs_{rms}]
600 $\leq f < 800$	75
800 $\leq f < 1000$	59
1000 $\leq f < 1400$	46
1400 $\leq f < 1600$	37
1600 $\leq f < 2000$	36
2000 $\leq f < 4000$	32
4000 $\leq f < 5000$	26
5000 $\leq f \leq 6000$	22

WSRF + WSPS stimulus random jitter typ

Frequency [MHz]	Random jitter typ [fs_{rms}]
600	56
885	40
1900	23
2400	19
5800	15

The following figure shows the typical measured stimulus random jitter as a function of output carrier frequency.



RFIM-6 stimulus EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB, using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.80%	0.5%
5.6 (-15 dBm to -2 dBm)	0.80%	0.55%

Optimizer: balanced, NoiseBandwidth=80 MHz

802.11ac 160MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.90%	0.65%
5.6 (-15 dBm to -2 dBm)	0.90%	0.7%

Optimizer: balanced, NoiseBandwidth=40 MHz

WSRF + WSPS stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.45%
5.6 (-15 dBm to -2 dBm)	0.55%	0.47%

Optimizer: balanced, NoiseBandwidth=80 MHz

802.11ac 160MHz

Frequency [GHz]	EVM
5.2 (-15 dBm to -2 dBm)	0.7% (max)
5.2 (-11 dBm to -2 dBm)	0.54% (typ)
5.2 (-15 dBm to -11 dBm)	0.62% (typ)
5.6 (-15 dBm to -2 dBm)	0.75% (max)
5.6 (-11 dBm to -2 dBm)	0.58% (typ)
5.6 (-15 dBm to -11 dBm)	0.66% (typ)

Optimizer: balanced, NoiseBandwidth=40 MHz

RFIM-6 stimulus ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	ACLR [dBc]	ACLR2 [dBc]
2 (-30 dBm to +4 dBm)	-65 (typ)	-72 (typ)
2 (-25 dBm to +1 dBm)	-63 (max) ^[4]	-70 (max) ^[4]

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 6dB

WSRF + WSPS stimulus ACLR

^[4] Some baseband frequency offset values might result in dominant spur yielding higher ACLR than spec. To avoid this, change the basebandFrequencyOffset value.

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	ACLR [dBc]	ACLR2 [dBc]
2 (-30 dBm to +4 dBm)	-68 (typ)	-72.5 (typ)
2 (-25 dBm to +1 dBm)	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 6dB

RFIM-6 stimulus noise floor

Stimulus noise floor

CW 0 dBm, optimizer mode setting: noiseFloor, measured at +40 MHz offset relative to carrier frequency

Carrier frequency range (GHz)	Noise-floor (dBm/Hz)
0.05 - 0.545	-138
0.545 - 3	-142
3 - 5	-140
5 - 6	-138

RFIM-6 stimulus spurs

Measured at 0 dBm stimulus power.

Note

Frequencies defined in *RFIM-6 stimulus harmonics* on page 13 are excluded from this specification.

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF-6 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-57
0.55	1.2	-65
1.2	1.7	-62
1.7	2	-60
2	2.05	-59
2.05	2.55	-60
2.55	3.1	-57
3.1	3.15	-53
3.15	3.4	-57
3.4	4	-54

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
4	4.1	-50
4.1	5.4	-53.5
5.4	5.45	-52
5.45	5.65	-54
5.65	5.9	-51
5.9	6	-54

Valid for spur frequency offset above +/- 40MHz

WSRF stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	2.5	-63
2.5	3.5	-60
3.5	4.5	-57
4.5	6	-50

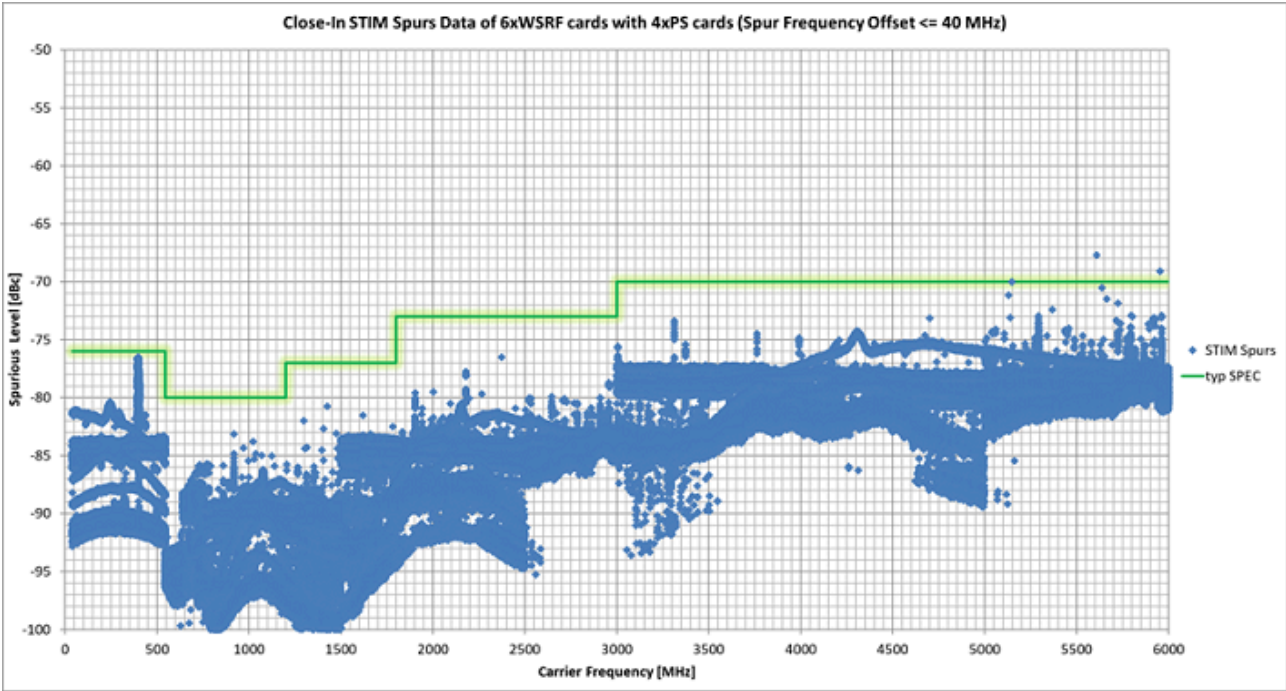
Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF + WSPS stimulus spurs max

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.01	6	-63

WSRF + WSPS stimulus spurs typ

Frequency from [GHz]	Frequency to [GHz]	Typ spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77
1.8	3	-73
3	6	-70



WSPS stimulus nominal spurious performance

Valid for spur frequency offset above +/- 40MHz

WSRF + WSPS stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63

Wave Scale RF-6 with RFIM-6 or RFIM-8 measure

RFIM-6 measure power

Applicable for high resolution ADC only.

Applicable for manual path selection and attenuation settings.

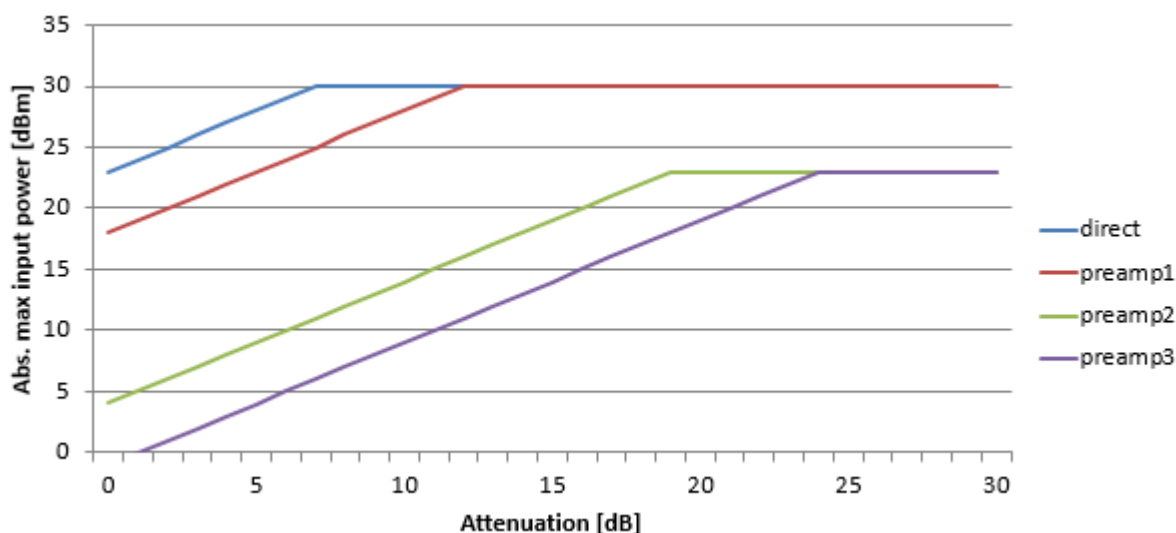
Note The specified maximum RF input power values below are only valid for the splitch in switch mode.

Maximum RF-6 input power to all ports

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
direct	0.01 to < 0.05	-1	3.0	29
	0.05 to < 1	3	6.9	30
	1 to < 2	5	8.8	30
	2 to < 3	7	10.2	30
	3 to < 4	7	10.7	30
	4 to < 5	7	11.2	30
	5 to 6	9	12.3	30
preamp1	0.01 to < 0.05	-10	-6.1	20
	0.05 to < 1	-6	-1.8	24
	1 to < 2	-3	0.3	27
	2 to < 3	-2	1.5	28
	3 to < 4	-1	2.6	29
	4 to < 5	-1	3.0	29
	5 to 6	0	3.5	30
preamp2	0.01 to < 0.1	-18	-14.7	12
	0.1 to < 0.2	-16	-12.1	14
	0.2 to < 1	-15	-12.0	15
	1 to < 2	-13	-9.9	17
	2 to < 3	-12	-8.5	18
	3 to < 4	-11	-7.7	19
	4 to < 5	-11	-7.2	19

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
	5 to 6	-11	-6.8	19
preamp3	0.01 to < 0.1	-28	-23.7	2
	0.1 to < 0.2	-24	-20.7	6
	0.2 to < 1	-24	-20.5	6
	1 to < 2	-22	-18.4	8
	2 to < 3	-21	-17.0	9
	3 to < 4	-20	-16.1	10
	4 to < 5	-19	-15.5	11
	5 to 6	-19	-15.2	11

Absolute maximum power damage level (dBm)



If the expected power value is larger than the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)", SmartTest will issue a warning.

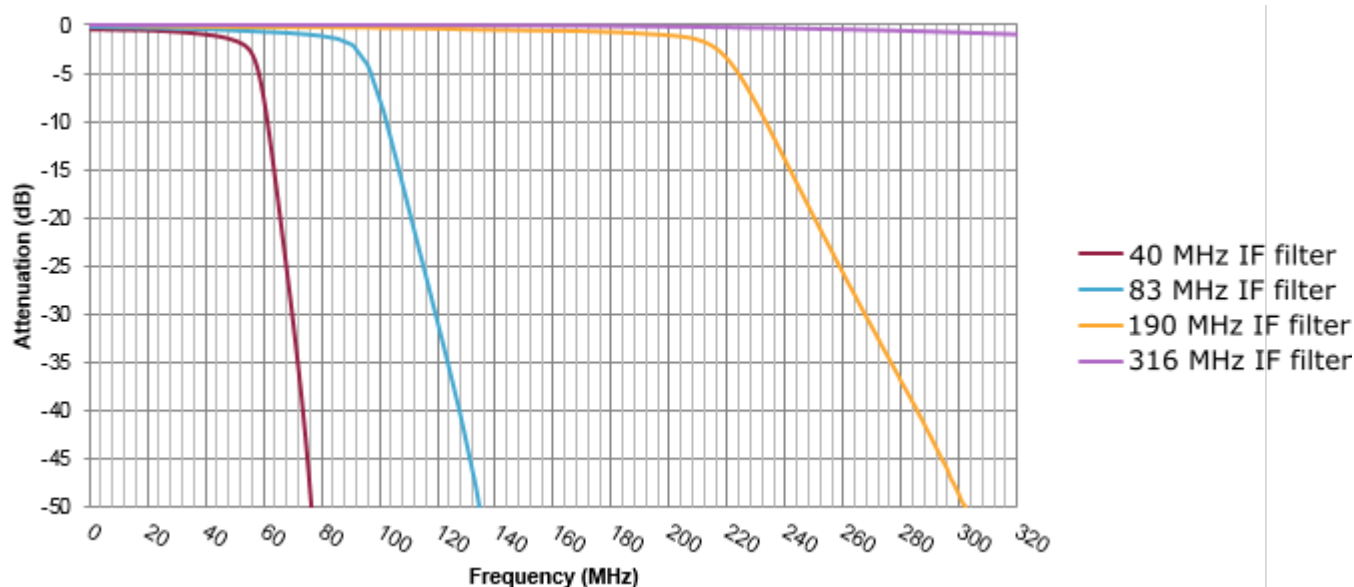
If the expected power value is larger than the "Absolute maximum power damage level(dBm)", SmartTest will issue an error.

Note

When setting *rfmeas actions*, the following aspects should be taken into account:

- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Absolute maximum power damage level (dBm)}$ on page 23 (see figure above).
- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Maximum measurable power with 0dB att, spec(dBm)} + \text{Attenuation(dB)}$ (see table above)
- If the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)" is higher than "Absolute maximum power damage level(dBm)", the latter has priority.

Typical IF filter response within the RF path



RFIM-6 measure accuracy up to 6 GHz

Measure power accuracy conditions:

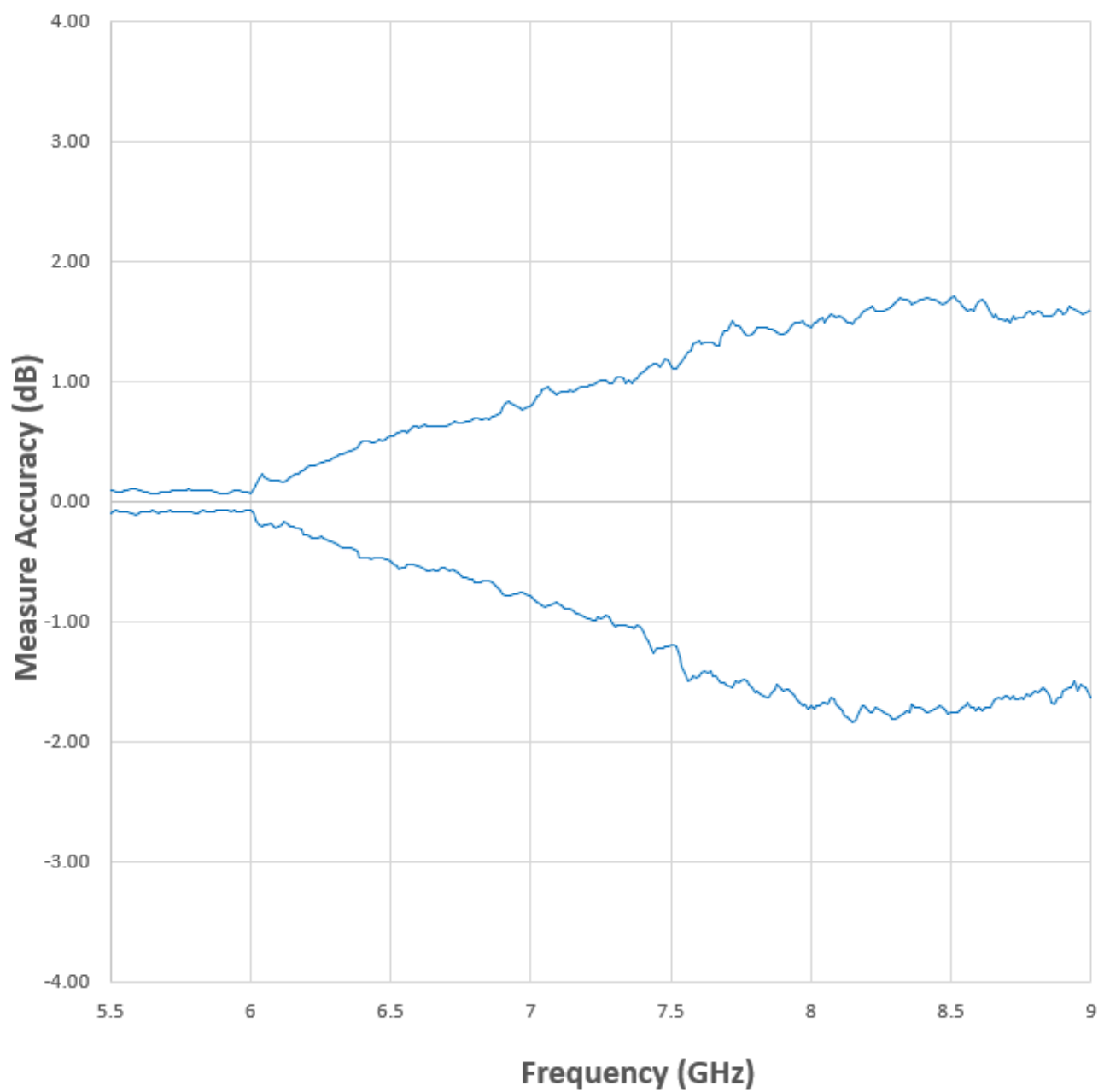
- All power levels are CW signals in a 50 ohm system unless noted.
- Standard deviation index $k=3$ (coverage factor of expanded uncertainty) unless specified differently.
- $\pm 0.5K$ relative water temperature stability after calibration.
- Level accuracy is valid for High Resolution ADC, only.
- To achieve specified level accuracy for very low power levels appropriate bandwidth and result averaging must be applied.

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
$0.050 \leq f < 0.545$	$+1 \leq L_v \leq +20$	± 0.38
	$-8 \leq L_v < +1$	± 0.38
	$-18 \leq L_v < -8$	± 0.38
	$-26 \leq L_v < -18$	± 0.38
	$L_v < -26$	± 0.38
$0.545 \leq f < 1.100$	$+3 \leq L_v \leq +20$	± 0.34
	$-6 \leq L_v < +3$	± 0.34
	$-16 \leq L_v < -6$	± 0.32
	$-24 \leq L_v < -16$	± 0.32
	$L_v < -24$	± 0.32
$1.100 \leq f < 1.990$	$+4 \leq L_v \leq +20$	± 0.36
	$-5 \leq L_v < +4$	± 0.34

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
	$-15 \leq L_v < -5$	± 0.34
	$-23 \leq L_v < -15$	± 0.34
	$L_v < -23$	± 0.34
$1.990 \leq f < 3.880$	$5 \leq L_v \leq +20$	± 0.40
	$-4 \leq L_v < 5$	± 0.40
	$-14 \leq L_v < -4$	± 0.40
	$-22 \leq L_v < -14$	± 0.40
	$L_v < -22$	± 0.40
$3.880 \leq f < 5.400$	$+6 \leq L_v \leq +20$	± 0.48
	$-3 \leq L_v < +6$	± 0.46
	$-13 \leq L_v < -3$	± 0.48
	$-21 \leq L_v < -13$	± 0.50
	$L_v < -21$	± 0.50
$5.400 \leq f \leq 6.000$	$+8 \leq L_v \leq +20$	± 0.42
	$-1 \leq L_v < +8$	± 0.40
	$-11 \leq L_v < -1$	± 0.40
	$-19 \leq L_v < -11$	± 0.40
	$L_v < -19$	± 0.40

RFIM-6 measure accuracy between 6 to 9 GHz

The measure accuracy for frequencies between 6 GHz and 9 GHz is shown below. This "measured" data represents 80% of the distribution of all measured curves.



WSRF measure accuracy between 6 to 9 GHz

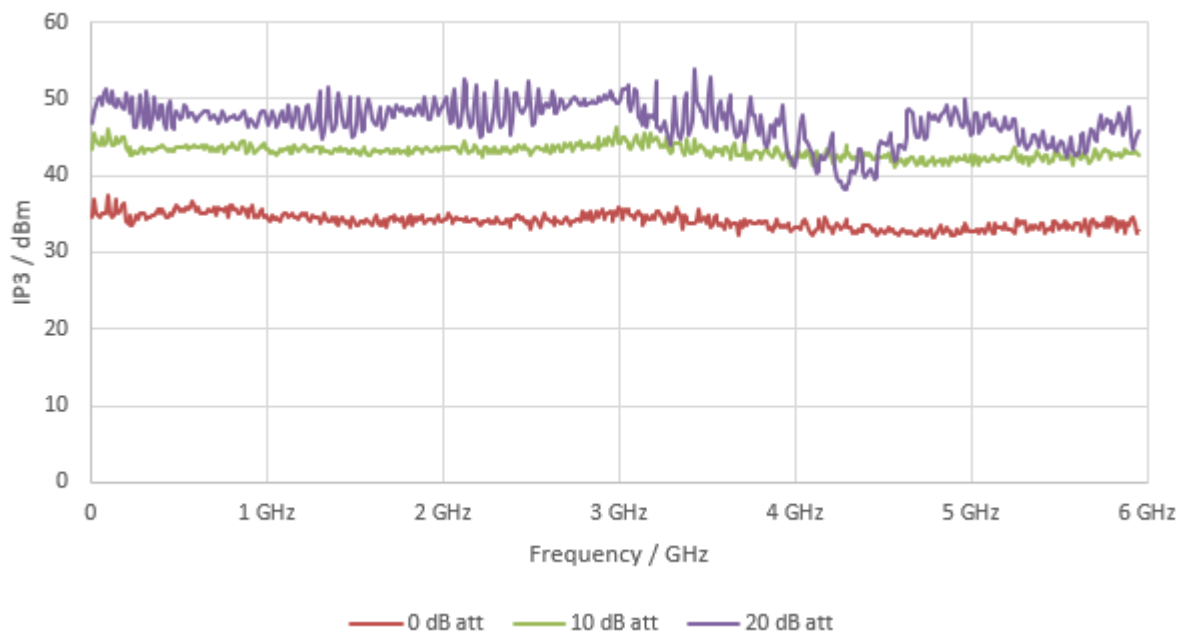
RFIM-6 measure two-tone intermodulation (IP3)

Conditions: HR ADC, 40 MHz filter, $f_{IF} = 0.533$ MHz, $\Delta f = 100$ kHz, -5 dBm tone power, with 0 dB receiver attenuation. IP3 increases with receiver attenuation.

Frequency: 50 MHz - 6 GHz

Measure Path	TOI (dBm)	
	Spec	Typ
direct	24	29
preamp1	15	19
preamp2	5	10

Measure Path	TOI (dBm)	
	Spec	Typ
preamp3	-4	1

Measured data

RFIM-6 measure two-tone intermodulation IP3 measured data

RFIM-6 measure EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM
2 (-40 dBm to +2 dBm)	0.80% spec
2 (-40 dBm to +14 dBm)	0.60% typ

802.11ac 80MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to +0 dBm)	0.80% spec
5.2 (-26 dBm to +10 dBm)	0.65% typ
5.6 (-26 dBm to +0 dBm)	0.90% spec
5.6 (-26 dBm to +10 dBm)	0.71% typ

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to +0 dBm)	1.00% spec

Frequency [GHz]	EVM
5.2 (-26 dBm to +10 dBm)	0.80% typ
5.6 (-26 dBm to +0 dBm)	1.10% spec
5.6 (-26 dBm to +10 dBm)	0.93% typ

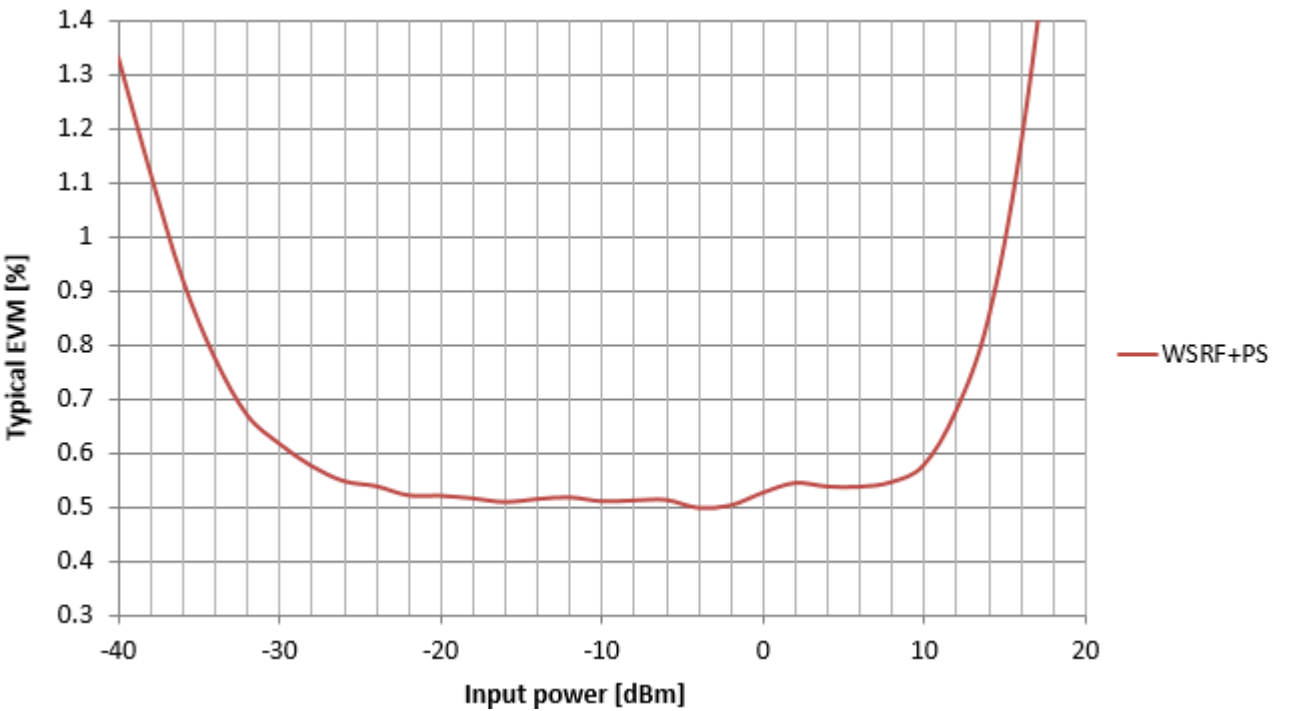
WSRF + WSPS measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM
2 (-40 dBm to +2 dBm)	0.8% (max)
2 (-40 dBm to +14 dBm)	0.54% (typ)

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to + 2 dBm)	1.0% (max)
5.2 (-26 dBm to +8 dBm)	0.54% (typ)



Wave Scale PS measure typ EVM 802.11ac 160MHz

RFIM-6 measure ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +6 dBm)	-63.0 (max)
2 (-16 dBm to +14 dBm) ^[5]	-65.7 (typ) ^[5]

WSRF + WSPS measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +6 dBm)	-67 (max)
2 (-16 dBm to +14 dBm)	-69 (typ)

RFIM-6 measure noise floor

Applicable for high resolution ADC only.

Measured at 0 dB receiver attenuation.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
direct	0.01 to < 0.05	-130	-136
	0.05 to < 1	-138	-143
	1 to < 2	-136	-141
	2 to < 3	-135	-141
	3 to < 4	-135	-140
	4 to < 5	-133	-139
	5 to 6	-132	-137
preamp1	0.01 to < 0.05	-139	-144
	0.05 to < 1	-145	-150
	1 to < 2	-144	-149
	2 to < 3	-143	-148
	3 to < 4	-142	-147
	4 to < 5	-141	-146
	5 to 6	-139	-144
preamp2	0.01 to < 0.1	-140	-145

^[5] Some IF value might result in dominant spur which yields to higher ACLR than spec. To avoid this, change IF value.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
	0.1 to < 0.2	-152	-157
	0.2 to < 1	-152	-157
	1 to < 2	-152	-157
	2 to < 3	-152	-156
	3 to < 4	-151	-156
	4 to < 5	-151	-155
	5 to 6	-150	-154
preamp3	0.01 to < 0.1	-142	-146
	0.1 to < 0.2	-153	-157
	0.2 to < 1	-155	-159
	1 to < 2	-155	-159
	2 to < 3	-155	-159
	3 to < 4	-155	-159
	4 to < 5	-154	-158
	5 to 6	-154	-157

RFIM-6 measure dynamic range

Applicable for high resolution ADC only.

Note

Dynamic range is the difference between noise floor (in dBm/Hz) and the maximum measurable power. It applies for 1 Hz measurement bandwidth and the dynamic range decreases with increasing bandwidth as $10\log(\text{bandwidth})$.

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
direct	0.01 to < 0.05	129	139
	0.05 to < 1	141	149.8
	1 to < 2	141	149.8
	2 to < 3	142	151.1
	3 to < 4	142	150.7
	4 to < 5	140	150.2
	5 to 6	141	149.3
preamp1	0.01 to < 0.05	129	137.9

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	0.05 to < 1	139	148.2
	1 to < 2	141	149.2
	2 to < 3	141	149.5
	3 to < 4	141	149.6
	4 to < 5	140	149
	5 to 6	139	147.4
preamp2	0.01 to < 0.1	122	130.3
	0.1 to < 0.2	136	144.9
	0.2 to < 1	137	144.9
	1 to < 2	139	147.1
	2 to < 3	140	147.4
	3 to < 4	140	148.2
	4 to < 5	140	147.7
	5 to 6	139	147.2
preamp3	0.01 to < 0.1	114	122.2
	0.1 to < 0.2	129	136.2
	0.2 to < 1	131	138.5
	1 to < 2	133	140.6
	2 to < 3	134	141.9
	3 to < 4	135	142.9
	4 to < 5	135	142.5
	5 to 6	135	141.7

RFIM-6 measure spurs

Measured at 0dBm stimulus power

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

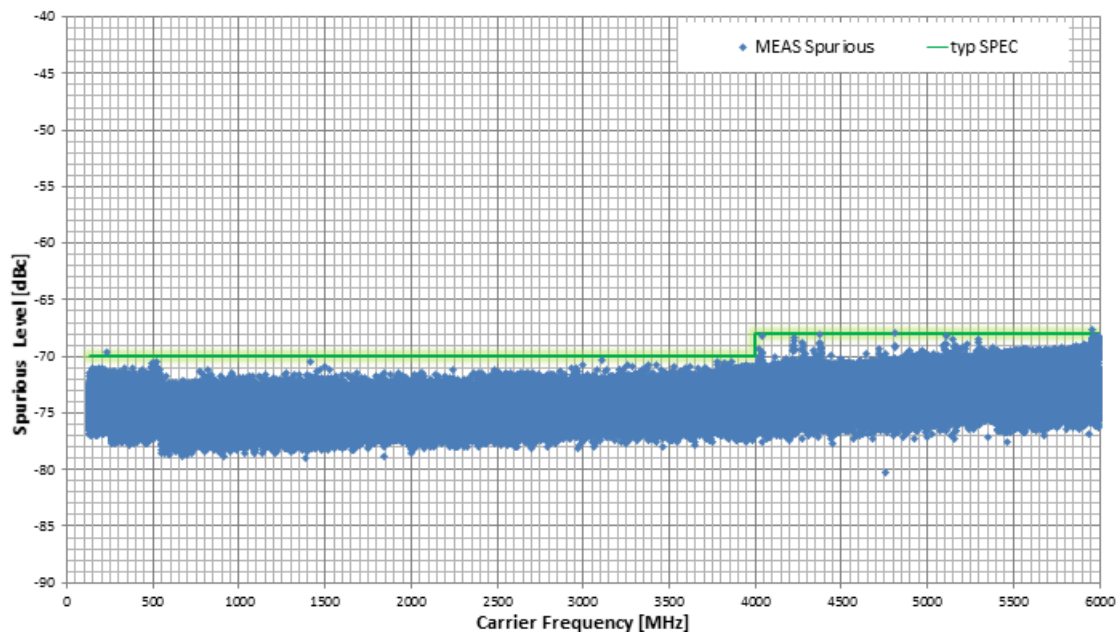
WSRF-6 measure spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
0.13	1	-67
1	1.2	-64

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
1.2	1.6	-61.5
1.6	1.7	-54
1.7	3.4	-56.5
3.4	4	-53
4	4.1	-51
4.1	5.15	-53
5.15	5.25	-52.5
5.25	5.9	-50.5
5.9	6	-53

WSRF + WSPS measure spurs

Input frequency from [GHz]	Input frequency to [GHz]	Typ spurs level (<0.01%) [dBc]
0.13	4	-70
4	6	-68



WS PS measure typical spurious performance

Wave Scale RF-6 with RFIM-6 or RFIM-8 terminate

RF ports that are turned-off, have an internal 50-Ohm termination to GND. In order to avoid thermal damage to the termination resistors, the following abs-max restrictions have to be observed per logical board (half of RFIM).

Number of ports in termination mode	Maximum power applied per port (dBm)
1 or 2	30
3 to 9	27
10 or more	26